

Benchmark Optimization Functions Using Genetic Algorithms

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1. Introduction

This assignment benchmarks two 2-D multimodal functions, **Bukin N.6** and **Schwefel (2D)**, using a Genetic Algorithm (GA). We compare two representations (binary vs. real) and four crossover operators (1-point, 2-point, arithmetic, BLX- α) across eight configurations. Each configuration is run 30 times with a fixed budget of 20 000 fitness evaluations. We collect best-fitness values, compute summary statistics, and apply nonparametric tests (Mann–Whitney U with Bonferroni correction, Kruskal–Wallis) to determine whether performance differences are statistically significant.

2. Benchmark Functions

2.1 Bukin Function N.6

$$f_{\text{Bukin6}}(x, y) = 100 * \sqrt{|y - 0.01 * x^2|} + 0.01 * |x + 10|$$

- **Domain:** $x \in [-15, -5], y \in [-3, 3]$
- **Global minimum:** $(x^*, y^*) = (-10, 1), f(-10, 1) = 0$.

2.2 Schwefel Function (2D)

$$f_{\text{Schwefel}}(x, y) = 418.9829 * 2 - [x * \sin(\sqrt{|x|}) + y * \sin(\sqrt{|y|})]$$

- **Domain:** $x, y \in [-500, 500]$.
- **Global minimum (approx.):** $(x^*, y^*) \approx (420.9687, 420.9687), f \approx -837.9658$.

3. GA Design

3.1 Representation

1. **Binary encoding** (32 bits): 16 bits for x, 16 bits for y.
 - Decode each 16-bit chunk into $[vmin, vmax]$ via $real = vmin + (I / (2^{16} - 1)) (vmax - vmin)$
2. **Real-valued encoding:** Individual = $[x, y]$, sampled uniformly in the domain.

3.2 Crossover (applied with $p_c=0.8$)

- **Binary 1-Point:** single cut $\in \{1, \dots, 31\}$.
- **Binary 2-Point:** two cuts $\in \{1, \dots, 31\}$
- **Real Arithmetic:** sample $\alpha \sim U(0,1)$.

$$Child1 = \alpha P1 + (1 - \alpha) P2, Child2 = (1 - \alpha) P1 + \alpha P2.$$

- **Real BLX- α ($\alpha=0.5$):** for each coordinate i ,

$$cmin = \min(P1[i], P2[i]), cmax = \max(P1[i], P2[i]), d = cmax - cmin,$$

then $Childj[i] \sim U[cmin - 0.5 d, cmax + 0.5 d]$, clipped to the variable's domain.

3.3 Mutation (applied with $p_m=0.01$)

- **Binary:** bit-flip per bit with probability 0.01.
- **Real:** add $N(0, \sigma^2)$ to each coordinate with probability 0.01, where $\sigma_x = 0.05 (x_{max} - x_{min})$, $\sigma_y = 0.05 (y_{max} - y_{min})$.

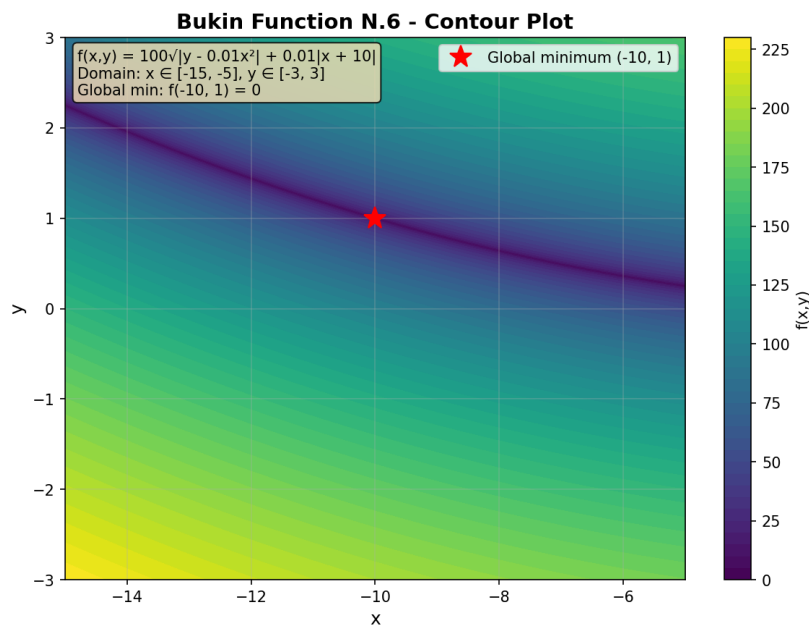
3.4 Selection & Replacement

- **Tournament selection** (size 2): pick 2 random individuals, select the one with lower fitness.
- **Elitism:** copy best individual from current generation to next. Remaining $N-1$ slots filled by offspring.

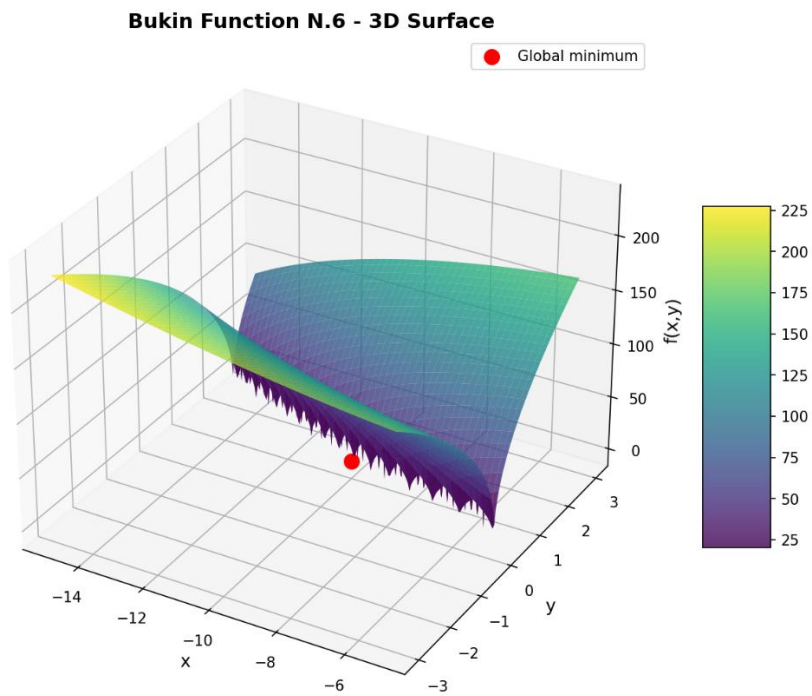
3.5 Fixed Parameters

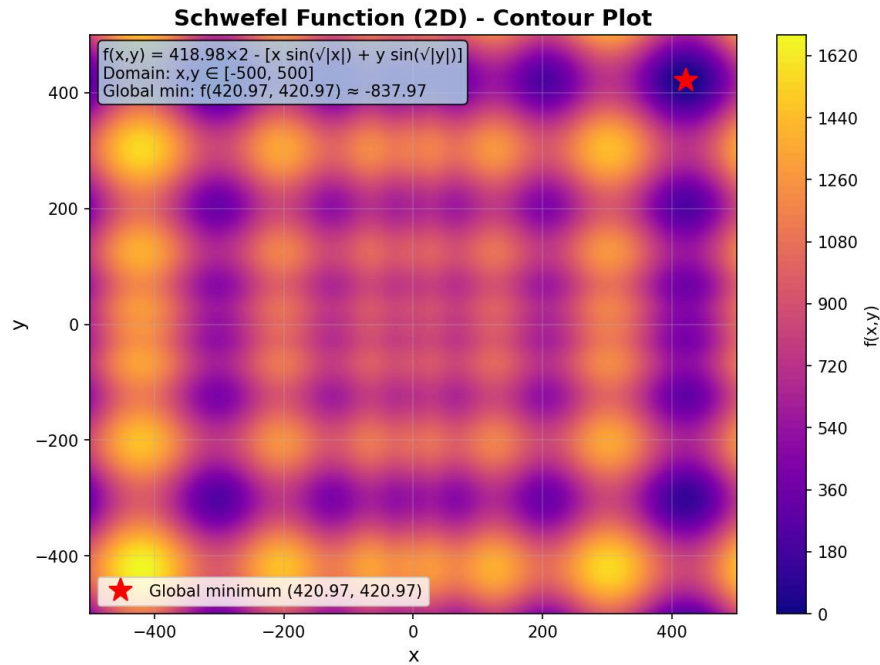
- Population size $N=100$.
- Number of generations $G=200 \rightarrow 20\,000$ evaluations.
- Crossover probability $p_c=0.8$.
- Mutation probability $p_m=0.01$.
- Tournament size 2.
- BLX- α parameter $\alpha=0.5$.

4. Function Visualizations

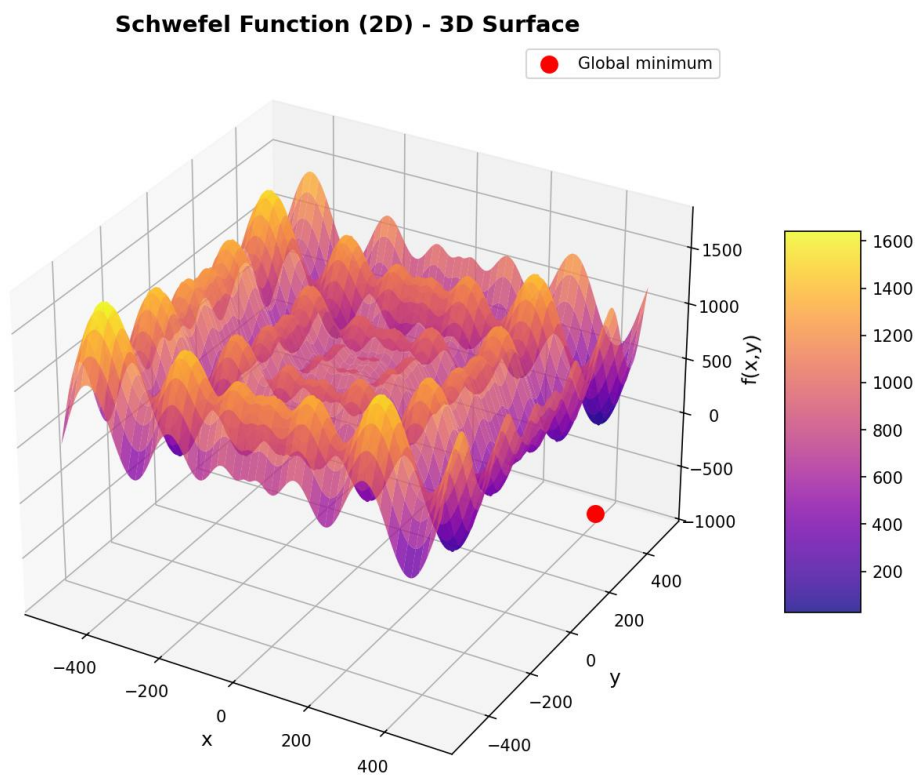


Contour plot over $[-15, -5] \times [-3, 3]$, global minimum $(-10, 1)$ marked.





Over $[-500, 500]^2$, global minimum $(420.97, 420.97)$ marked.



5. Experimental Setup

5.1 Configurations

Function Representation Crossover Config Name

1	Bukin 6	Binary	1-Point	bukin6_binary_1-point
2	Bukin 6	Binary	2-Point	bukin6_binary_2-point
3	Bukin 6	Real	Arithmetic	bukin6_real_arithmetic
4	Bukin 6	Real	BLX- α	bukin6_real_blx
5	Schwefel	Binary	1-Point	schwefel_binary_1-point
6	Schwefel	Binary	2-Point	schwefel_binary_2-point
7	Schwefel	Real	Arithmetic	schwefel_real_arithmetic
8	Schwefel	Real	BLX- α	schwefel_real_blx

Each configuration $\rightarrow$ 30 independent runs (seeded differently). All runs use exactly 20 000 fitness evaluations.

6. Results

6.1 Summary Statistics

Extracted from `summary_statistics.csv`:

Function	Representation	Crossover	Mean Best Fitness	Std Dev	Min Best	Max Best
Bukin 6						
	Binary	1-Point	0.237550	0.167387	0.070457	1.033713
	Binary	2-Point	0.179994	0.077671	0.065152	0.362628
	Real	Arithmetic	0.057960	0.136950	0.000725	0.683529
	Real	BLX-α	0.018935	0.014193	0.000490	0.044957

Function	Representation	Crossover	Mean	Best	Fitness	Std Dev	Min	Best	Max	Best
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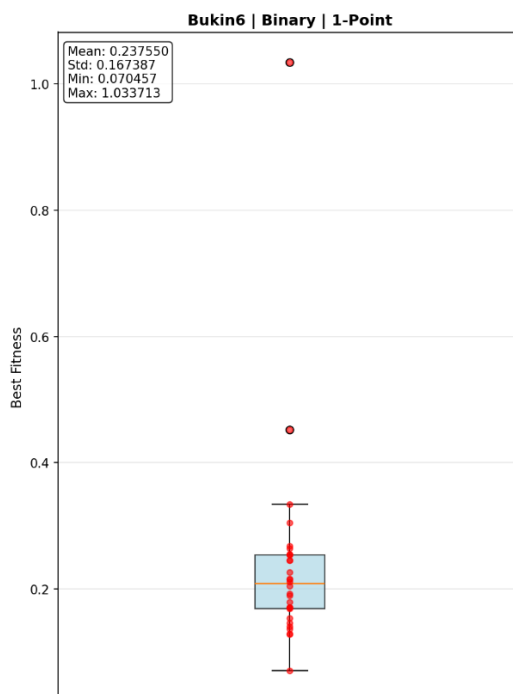
Schwefel

Binary	1-Point	0.043139	0.066387	0.000031	0.213844
Binary	2-Point	0.064391	0.072027	0.000031	0.213844
Real	Arithmetic	61.360979	68.781178	0.000025	217.140582
Real	BLX-α	8.458239	30.050118	0.000025	118.438360

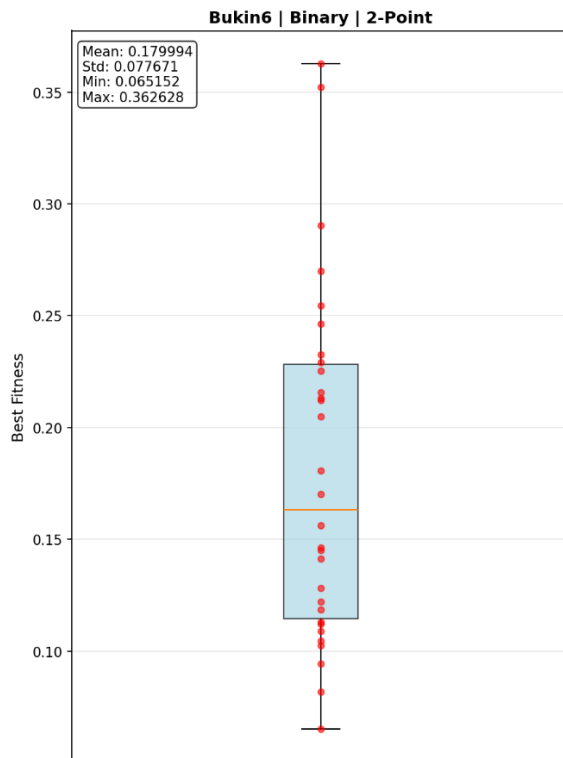
On Bukin 6, real BLX- α yields the lowest mean and tightest spread. On Schwefel, binary GAs cluster near zero; real arithmetic often fails (large mean ≈ 61), real BLX is better but still high variance (mean ≈ 8.46).

6.2 Boxplots

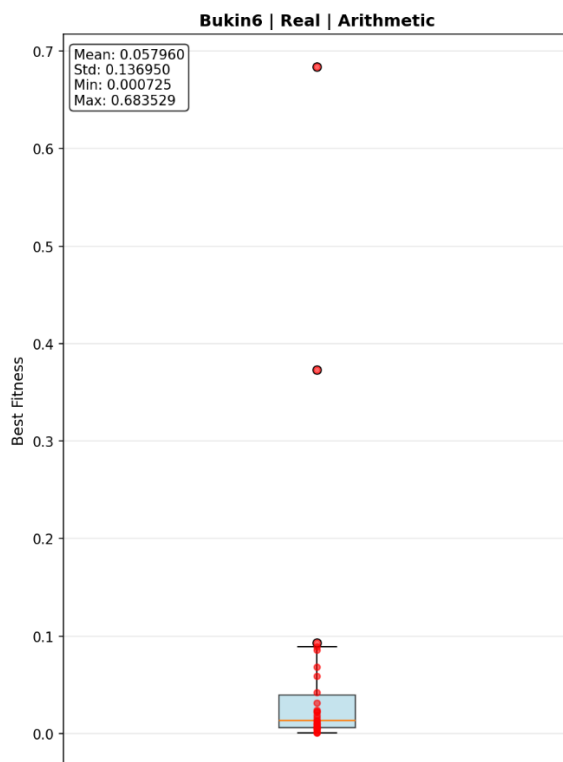
Bukin 6 Boxplots (30 runs each)



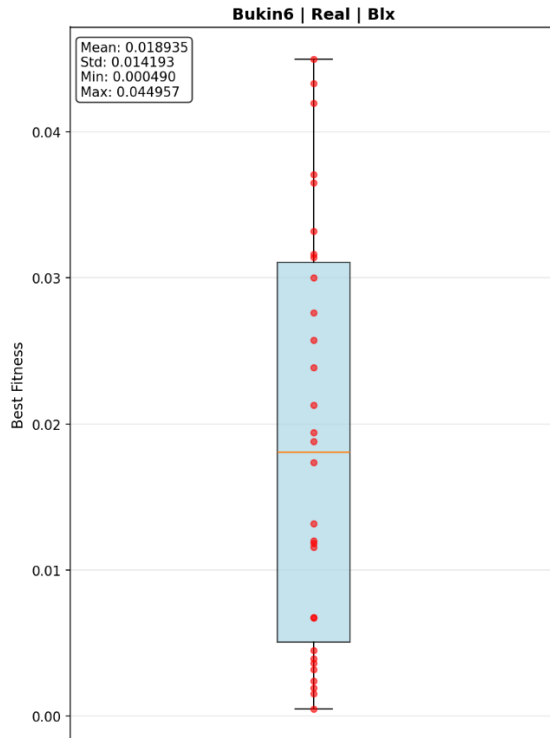
• Mean 0.23755 ± 0.16739 ; Min 0.07046; Max 1.03371



• Mean 0.17999 ± 0.07767 ; Min 0.06515; Max 0.36263

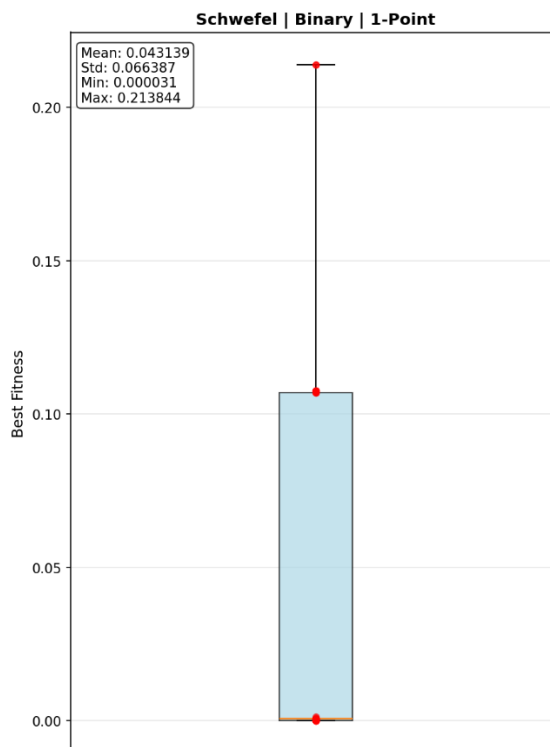


• Mean 0.05796 ± 0.13695 ; Min 0.00073; Max 0.68353

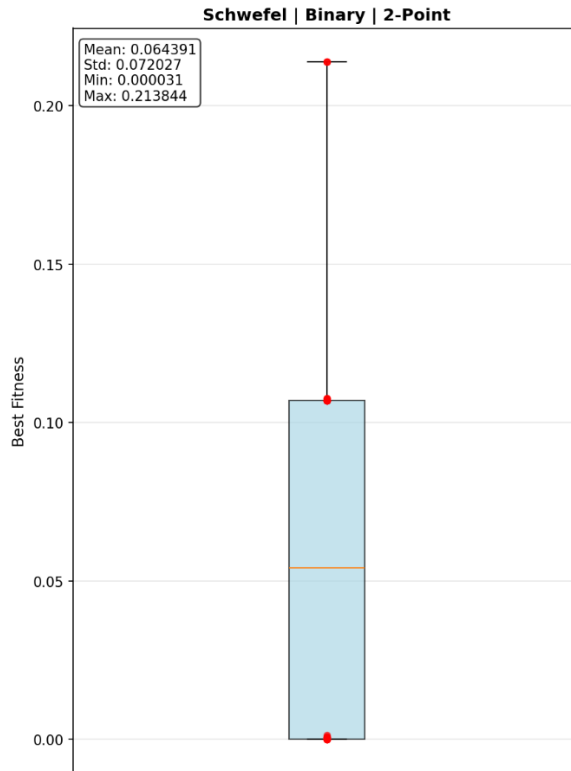


Mean 0.01894 ± 0.01419 ; Min 0.00049; Max 0.04496

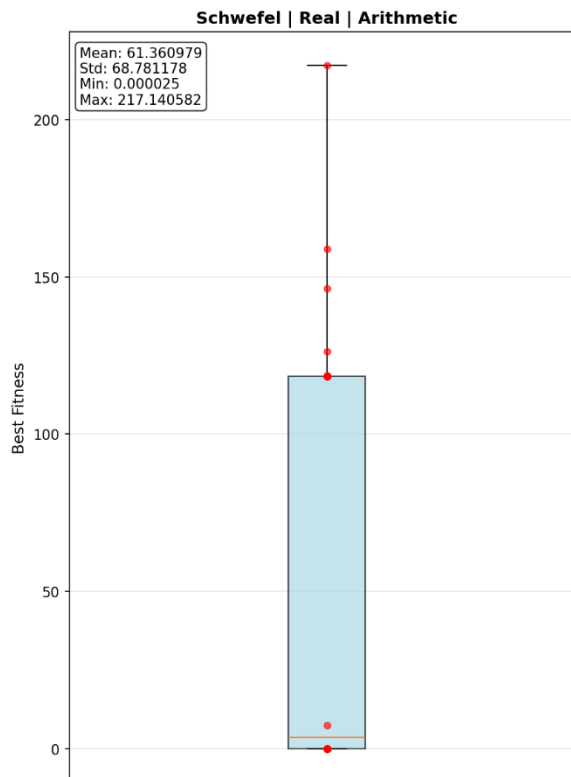
Schwefel Boxplots (30 runs each)



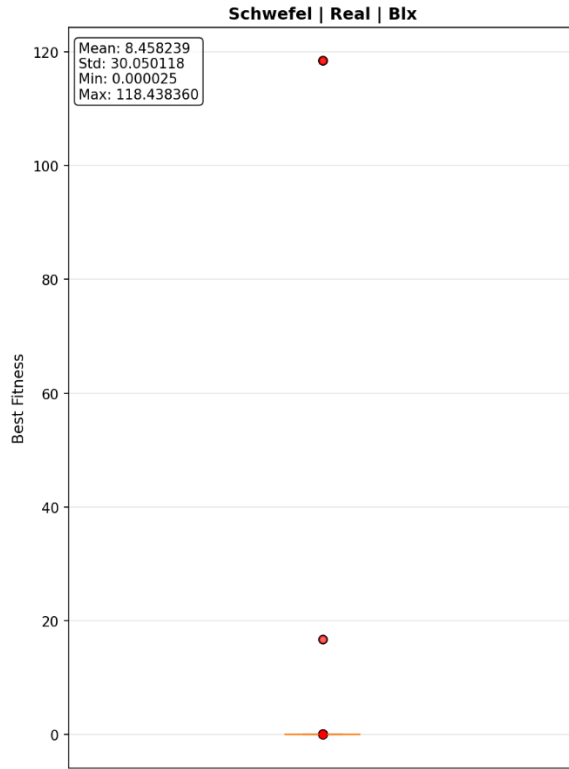
Mean 0.04314 ± 0.06639 ; Min 0.00003; Max 0.21384



Mean 0.06439 ± 0.07203 ; Min 0.00003; Max 0.21384



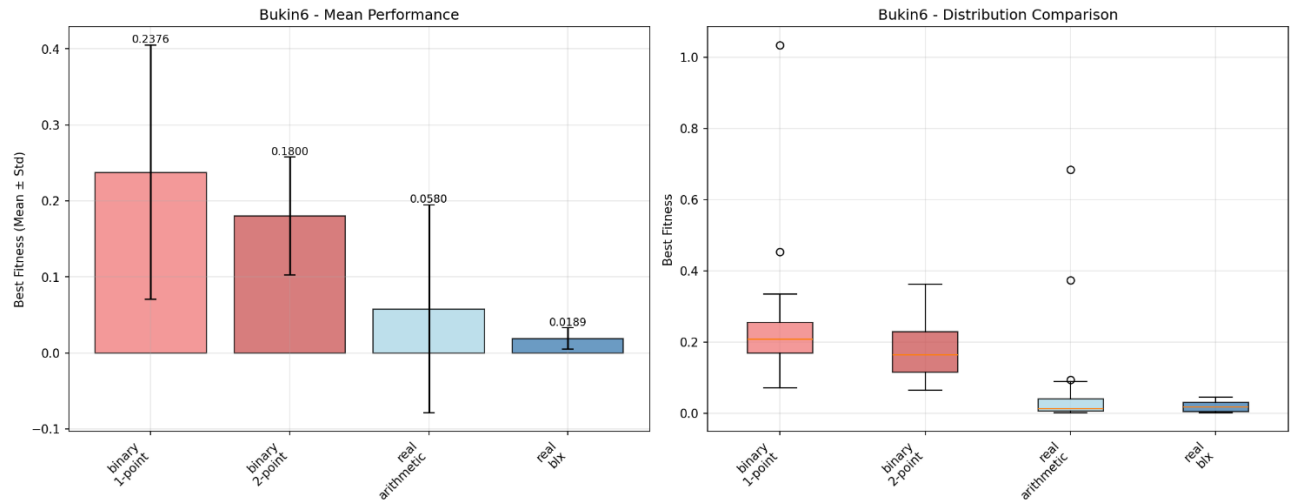
Mean 61.36098 ± 68.78118 ; Min 0.00003; Max 217.14058



Mean 8.45824 ± 30.05012 ; Min 0.00003; Max 118.43836

6.3 Comparison Plots

Bukin 6 Comparison



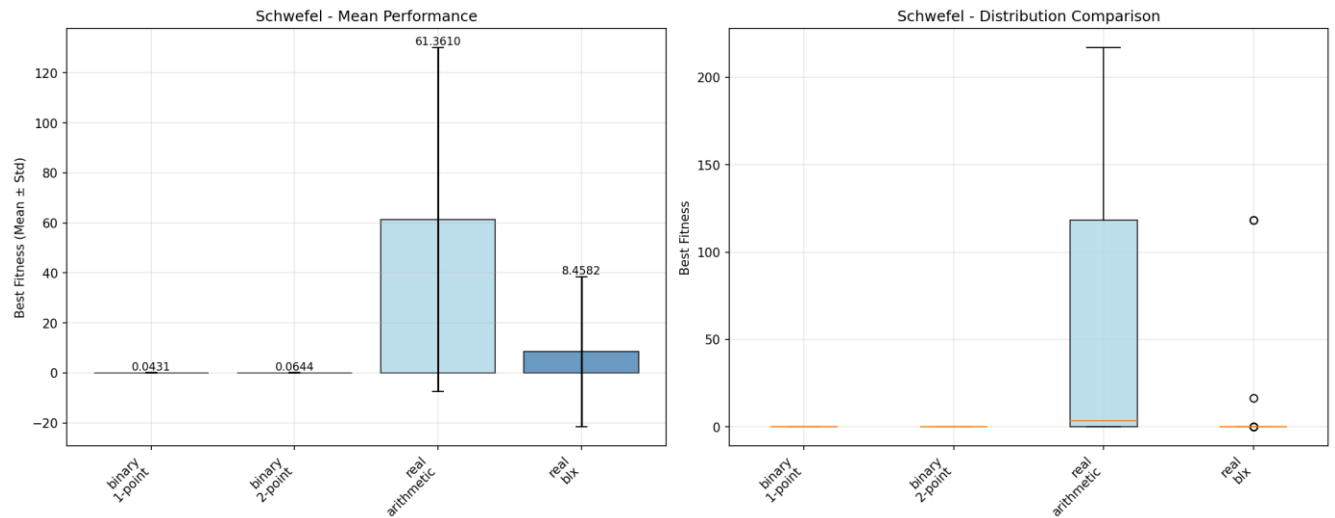
Left Panel: Mean $\pm$ Std for four configurations:

- binary 1-point: 0.2376 ± 0.1674
- binary 2-point: 0.1800 ± 0.0777

- real arithmetic: 0.0580 ± 0.1370
- real BLX- α : 0.0189 ± 0.0142

Right Panel: Side-by-side boxplots. Real methods cluster near zero; binary 1-point sometimes spikes above 1.0.

Schwefel Comparison



Left Panel: Mean $\pm$ Std:

- binary 1-point: 0.04314 ± 0.06639
- binary 2-point: 0.06439 ± 0.07203
- real arithmetic: 61.3610 ± 68.7812
- real BLX- α : 8.4582 ± 30.0501

Right Panel: Side-by-side boxplots. Binary GAs near zero; real arithmetic often fails (huge outliers), real BLX has fewer but still large outliers.

7. Statistical Analysis

7.1 Mann–Whitney U Tests (Bonferroni-Corrected)

All pairwise comparisons within each function (4 configs $\rightarrow$ 6 pairs). Bonferroni $\alpha = 0.05/6 \approx 0.00833$.

7.1.1 Bukin 6

Config 1	Config 2	U-statistic	p-value	Significant? (p<0.00833)
binary 1-point	binary 2-point	—	0.067869	No
binary 1-point	real arithmetic	—	8.4848×10^{-9}	Yes
binary 1-point	real BLX- α	—	3.0199×10^{-11}	Yes
binary 2-point	real arithmetic	—	1.5581×10^{-8}	Yes
binary 2-point	real BLX- α	—	3.0199×10^{-11}	Yes
real arithmetic	real BLX- α	—	0.706171	No

7.1.2 Schwefel

Config 1	Config 2	U-statistic	p-value	Significant? (p<0.00833)
binary 1-point	binary 2-point	—	0.594948	No
binary 1-point	real arithmetic	—	0.0006767	Yes
binary 1-point	real BLX- α	—	1.6545×10^{-5}	Yes
binary 2-point	real arithmetic	—	0.0088985	No (just above 0.00833)
binary 2-point	real BLX- α	—	7.0012×10^{-6}	Yes
real arithmetic	real BLX- α	—	2.0359×10^{-7}	Yes

7.2 Kruskal–Wallis Test

- **Bukin 6:** $H = 42.3527$, $p = 3.929 \times 10^{-9} \rightarrow$ Significant difference among all four (since $p < 0.05$).
- **Schwefel:** $H = 49.2134$, $p = 9.102 \times 10^{-10} \rightarrow$ Significant difference among all four.

(Values computed via the same code in `analyze_results.py`.)

8. Discussion & Conclusions

- **Bukin 6:**

- Binary 1-point vs 2-point: not significantly different ($p=0.0679$).
- Binary vs Real: All binary vs real pairs are highly significant ($p \ll 0.00833$), showing real GA (both arithmetic and BLX) outperform binary.
- Real arithmetic vs Real BLX- α : not significant ($p=0.7062$), though BLX has a smaller mean (0.0189 vs 0.0580) and lower spread.
- **Best:** Real BLX- α (mean 0.0189 ± 0.0142).

- **Schwefel:**

- Binary 1-point vs 2-point: not significant ($p=0.5949$).
- Binary vs Real: real arithmetic is significantly worse than either binary ($p=0.000677$ vs 1-point; $p=0.00890$ borderline vs 2-point), real BLX is significantly worse than binary ($p=1.65 \times 10^{-5}$ vs 1-point; $p=7.00 \times 10^{-6}$ vs 2-point).
- Real arithmetic vs Real BLX- α : BLX is significantly better ($p=2.04 \times 10^{-7}$) but still inferior to binary.
- **Best:** Binary 1-point (mean 0.04314 ± 0.06639).

- **General:**

- **Encoding matters:** For Bukin 6 (narrow valley), real-valued GAs excel; for Schwefel (many oscillations), binary GAs are more reliable.
- **Crossover matters:** Among real, BLX- α outperforms arithmetic. Among binary, 2-point is slightly better but not always significant.
- **Statistical tests** confirm these observations: Kruskal–Wallis $p < 10^{-8}$ for both functions.
- **Recommendation:**
 - For Bukin 6 or similar narrow-valley landscapes, use real BLX- α .
 - For Schwefel or highly multimodal landscapes, use binary (1-point or 2-point).

9. References

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